

Week 3 · Laboratory Solutions

- These are **reference** answers, not the only correct ones. The checker tests the physics, not the diagram.
- Read them after attempting the problem. The scripts explain what to build **and why the alternatives were rejected**.

Building and checking

```
cd lectures/W03_simulink/solutions
W03_S1_speed_loop      % builds W03_S1.slx – all three problems in one model
W03_S_expected         % regenerates the expected-result figures

W03_check(1,'W03_S1')
W03_check(2,'W03_S1')
W03_check(3,'W03_S1')
```

All three pass, and `check_overlaps('W03_S1')` is **0**. The output below is the measured one.

```
W03 problem 1
u_ss at X = 50 N           0.6447 (expected  0.6447 +- 0.005 m/s) PASS
u_ss at X = 100 N          1.2894 (expected  1.2894 +- 0.005 m/s) PASS
u_ss at X = 200 N          2.5788 (expected  2.5788 +- 0.005 m/s) PASS

W03 problem 2
u_ss at Kp = 100           0.8448 (expected  0.8448 +- 0.005 m/s) PASS
u_ss at Kp = 500           1.2986 (expected  1.2986 +- 0.005 m/s) PASS
u_ss at Kp = 2000          1.4440 (expected  1.4440 +- 0.005 m/s) PASS

W03 problem 3
steady speed with PI        1.5000 (expected  1.5000 +- 0.005 m/s) PASS
steady error with PI        -0.0000 (expected  0.0000 +- 0.005 m/s) PASS
overshoot with PI           1.1682 [%] (P alone had none)
inside 2 % after (section E: 1.30 s) 1.2800 (expected  1.3000 +- 0.1 s) PASS
```

Why one model and not three

The three problems differ only in which parts of the loop are switched on:

	<code>loop_closed</code>	K_i	
Problem 1	0	—	X comes straight from <code>X_open</code>
Problem 2	1	0	proportional only
Problem 3	1	> 0	proportional plus integral

Three separate models would hide the one fact the week is about: **the plant never changed and the thrust map never changed**. Every difference in the result came from the controller. The lecture's models hold every gain as a workspace variable for the same reason: a section script sweeps a gain without rebuilding anything.

The three decisions worth defending

	The choice	Why the alternative is worse
Discrete integrator	Discrete-Time Integrator at sample time <code>h</code>	A continuous integrator in a fixed-step model gives a solver-order mismatch. It does not raise an error — it produces a slow drift that looks like a physical effect
Feedback is u, state 1	one Selector, then one line	Feeding back $\sqrt{u^2 + v^2}$ agrees here and disagrees in Week 5. A loop written against the wrong signal keeps working until exactly the moment it matters
Feedback is a Goto/From tag	<code>u_fb</code>	Drawn as a line it crosses the whole model backwards and lands on the forward path. <code>check_overlaps</code> reported exactly one overlapping pair when it was a line

The layout trap this model walked into

All three inputs of a Switch block arrive on its **left edge**. Autorouting therefore gives them the same vertical lane, and two of them end up drawn on top of each other — `check_overlaps` found precisely that at $x = 425$.

The fix is `_tools/lane_line.m`, which forces a chosen vertical lane per signal:

```
lane_line mdl, 'loop_closed', 1, 'loop', 2, 424);  
lane_line mdl, 'X_open',      1, 'loop', 3, 412);
```

This is worth knowing before building any model with a Switch or a multi-input Mux, which is most of the models from Week 5 onward.